

NIR Photovoltaic Power Delivery for Implantable Devices

Executive Technology Brief

Domain: Optics + Photonics + Power Electronics

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Implantable sensors need power without batteries or leads

Mechanism: External 940nm NIR source penetrates tissue; implanted GaAs PV cell converts to electrical energy

Architecture: External 940nm LED → scalp/skull tissue → implanted GaAs PV cell → power conditioning → sensor/load

Governing Model: Near-infrared photovoltaic power delivery through tissue for implantable shunt sensor. NIR light from external source penetrates tissue; implanted PV cell converts to electrical energy.

Key Equations:

Beer-Lambert: $I(d) = I_0 * \exp(-\mu_{eff} * d)$ [P-16 NIR tissue attenuation through scalp/skull to implanted PV cell]

$P_{electrical} = P_{optical} * \eta_{PV}$ [P-16 PV conversion at implanted catheter PV cell]

$\mu_{eff} \sim 1-10 \text{ cm}^{-1}$ in tissue at 800-1000 nm (EXTERNAL_PRECEDENT)

Kill Condition: Tissue attenuation at 5cm depth prevents sufficient power delivery (>1mW/cm² required at PV surface)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.